

tem security policy administrator.

The concept of a trusted path can be generalised to include interactions beyond just those between trusted software and users. The TNI introduces the concept of a trusted channel for communication between trusted software on different network components. More generally, a mechanism that guarantees a mutually authenticated channel, or protected path, is necessary to ensure that critical system functions are not being spoofed. Although a protected path mechanism for local communications could be constructed in application space without direct authentication support in the operating system, it is preferable for an operating system to provide its own protected path mechanism since such a mechanism will be simpler to assure and is likely to be more efficient.

Most mainstream commercial operating systems are utterly lacking in their support for either a trusted path mechanism or a protected path mechanism. Microsoft Windows NT does provide a trusted path for a small set of functions such as login authentication and password changing but lacks support for extending the trusted path mechanism to other trusted applications. For local communications, NT does provide servers with the identity of their clients; however, it does not provide the server identity to the client.

General Examples

Without operating system support for mandatory security and trusted path, application space mechanisms for access control and cryptography cannot be implemented securely.

Access Control: An application-space access control mechanism may be decomposed into an enforcer component and a decider component. When a subject attempts to access an object protected by the mechanism, the enforcer component must invoke the decider component, supplying it with the proper input parameters for the policy decision, and must enforce the returned decision. A common example of the required input parameters is the security attributes of the subject and the object. The decider component may also consult other external sources in order to make the policy decision. For example, it may use an external policy